Lecture101 — Foundations of Data Science

Overview

Foundations of Data Science, Artificial Intelligence, and Emerging Paradigms PART I — Artificial Intelligence and Data Science Foundations 1. Artificial Intelligence (AI): The Umbrella Field Artificial Intelligence (AI) is a broad field of computer science focused on building systems capable of performing tasks that typically require human intelligence, such as reasoning, learning, perception, language understanding, and decision-making.

Learning Objectives

• Understand core concepts and terminology in Data Science.

• Familiarize with common data analysis workflows and tools.

• Apply basic statistical and machine learning techniques to datasets.

Prerequisites

Basic programming (Python) and introductory statistics recommended.

Weekly Topics / Detailed Content

Foundations of Data Science, Artificial Intelligence, and Emerging Paradigms

PART I — Artificial Intelligence and Data Science Foundations

1. Artificial Intelligence (AI): The Umbrella Field

Artificial Intelligence (AI) is a broad field of computer science focused on building systems capable of performing tasks that typically require human intelligence, such as reasoning, learning, perception, language understanding, and decision-making.

AI includes several major subfields:

Machine Learning

Deep Learning

Data Mining

Generative AI

Agentic AI

2. Machine Learning and Deep Learning

2.1 Machine Learning (ML)

Machine Learning enables systems to learn patterns from data and make predictions or decisions without explicit rule-based programming.

Common ML Tasks

Classification

Regression

Clustering

2.2 Deep Learning (DL)

Deep Learning is a specialized branch of ML that uses multi-layer neural networks to learn hierarchical feature representations from large and complex datasets.

Deep learning powers:

Large Language Models (LLMs)

Image generation systems

Speech recognition

Modern generative models

3. Data Mining vs Data Science

3.1 Data Mining

Data Mining focuses on discovering meaningful patterns and relationships from large datasets using statistical and machine learning techniques.

Primary emphasis:

Pattern discovery

Knowledge extraction

Interpretation

3.2 Data Science

Data Science is an interdisciplinary field integrating:

Statistics

Data Mining

Machine Learning

Data Engineering

Domain Knowledge

Goal: Extract actionable insights, predictions, and decisions from data.

Unlike traditional data mining, Data Science emphasizes:

End-to-end pipelines

Automation

Model deployment

Decision support

✓ Data mining is one component within the broader Data Science lifecycle.

PART II — The Data Science Lifecycle

4. Extended Knowledge Discovery Process

Problem / Business Understanding

Data Collection

Data Cleaning & Preprocessing

Exploratory Data Analysis (EDA)

Feature Engineering

Modeling (ML / DL)

Evaluation & Validation

Deployment & Monitoring

This lifecycle transforms raw data into deployed, value-generating systems.

PART III — Data Foundations

5. Core Data Concepts

5.1 Dataset

A structured, semi-structured, or unstructured collection of observations representing real-world entities.

5.2 Subject (Entity of Analysis)

Defines what the dataset represents.

Examples:

Students → Educational analytics

Patients → Healthcare analytics

Customers → Business analytics

5.3 Observation / Instance

Each row represents one entity or event

Also called: instance, record, sample

✓ Correct identification of subject and observation unit is essential for valid modeling.

6. Structure of Data

6.1 Structured Data

Tabular (rows × columns)

Easily modeled using classical ML

Examples:

CSV files

Excel sheets

Relational databases

6.2 Semi-Structured Data

Flexible schema

Examples:

JSON

XML

Logs

API responses

6.3 Unstructured Data

No fixed schema

Requires feature extraction

Examples:

Text (NLP)

Images (Computer Vision)

Audio / Video

✓ Modern AI heavily focuses on semi-structured and unstructured data.

7. Features and Variables

7.1 Features (Independent Variables)

Characteristics describing the subject.

Example (Student dataset):

Attendance rate

Prior GPA

LMS activity time

7.2 Target Variable (Label)

The outcome to be predicted.

Examples:

Final GPA

Pass / Fail

Dropout risk

✓ Feature engineering is a core Data Science skill.

8. Measurement Scales and Data Types

Understanding data types is essential for:

Algorithm selection

Distance computation

Feature encoding

8.1 Nominal (Categorical – Unordered)

No intrinsic order

Requires encoding (e.g., One-Hot Encoding).

8.2 Ordinal (Categorical – Ordered)

Order matters

Distance does not

Order must be preserved during encoding.

8.3 Numerical (Quantitative)

Discrete – Countable integersContinuous – Real-valued measurements

Required for:

Regression

Distance-based algorithms

Neural networks

9. Labeled vs Unlabeled Data

Labeled Data

Features + known output

Used in supervised learning

Unlabeled Data

No target variable

Used in unsupervised learning

✓ Semi-supervised learning combines both.

10. Sources of Data

Primary Data

Collected directly

High control, higher cost

Secondary Data

Pre-existing datasets

Widely used in research

PART IV — Analytics Levels in Data Science

11. Descriptive / Exploratory Analytics

Question: What happened?Uses summary statistics, visualization, and pattern discovery.

12. Predictive Analytics

Question: What will happen?Uses classification, regression, forecasting with ML/DL.

13. Prescriptive Analytics

Question: What should we do?Uses optimization, recommendation systems, and decision support.

✓ Highest level of analytics maturity.

PART V — Generative AI

14. What is Generative AI?

Generative AI refers to models that learn the underlying probability distribution of data and generate new, synthetic content that resembles training data.

Unlike predictive models, generative models create new data instances.

15. Core Characteristics

Learns data distributions

Produces novel outputs (text, images, audio, code, video)

Probabilistic and stochastic

Trained on large-scale datasets

16. Common Generative Models

Autoregressive Models (e.g., LLMs)

Variational Autoencoders (VAEs)

Generative Adversarial Networks (GANs)

Diffusion Models

17. Generative AI in Practice: AI Coding Assistants

One practical and widely adopted example of Generative AI is GitHub Copilot.

What is GitHub Copilot?

GitHub Copilot is an AI-powered code generation assistant integrated into development environments (e.g., VS Code). It is trained on large-scale code and natural language data and uses deep learning models (LLMs) to:

Autocomplete code

Generate functions from natural language prompts

Suggest algorithms

Create test cases

Explain existing code

Why It Matters in Data Science

For Data Scientists, GitHub Copilot can:

Generate data preprocessing pipelines

Suggest feature engineering code

Produce visualization scripts

Assist with model implementation (e.g., regression, classification)

Speed up experimentation and prototyping

✓ Copilot is an example of Generative AI applied to software development workflows.

PART VI — Agentic AI

18. What is Agentic AI?

Agentic AI refers to autonomous AI systems that:

Perceive environments

Reason about goals

Plan actions

Execute decisions

Adapt over time

Agentic systems emphasize autonomy and goal-directed behavior.

19. Core Components of an AI Agent

Perception – Observing inputs/environment

Reasoning – Interpreting and deciding

Planning – Sequencing actions

Action – Executing via tools/APIs

Memory – Retaining context

20. Applications of Agentic AI

Autonomous research assistants

Intelligent tutoring systems

Workflow automation agents

Academic advising systems

Decision-support systems

Agentic systems may use generative models (like those powering GitHub Copilot) as their reasoning engine, but extend them with planning, memory, and tool-use capabilities.

PART VII — Integration and Comparison

21. Generative AI vs Agentic AI

Aspect

Generative AI

Agentic AI

Primary Focus

Content generation

Goal-driven action

Core Capability

Generate data

Decide and act

Autonomy

Limited

High

Interaction

Prompt-based

Environment-based

Memory

Often short-term

Persistent

Output

Text, images, code

Actions, plans, decisions

✓ Generative AI often acts as the cognitive engine within agentic systems.

22. Architectural Perspectives

22.1 Generative Model Flow

Training Data→ Deep Neural Network→ Learned Probability Distribution→ Generated Output

22.2 Agentic Feedback Loop

Environment↔ AI Agent (Perception → Reasoning → Planning → Action → Memory)↔ Environment

22.3 Combined Architecture

Agent→ Generative Model (LLM)→ Planning Logic→ Tool Execution→ Environment

Modern intelligent systems increasingly integrate:

Data-centric modeling

Generative intelligence

Autonomous agent architectures

PART VIII — Ethical and Practical Considerations

Bias and fairness

Hallucinations and reliability

Over-reliance on AI-generated code (e.g., Copilot suggestions must be reviewed)

Data privacy and security

Responsible deployment in education and research

Final Integrated Summary

Data Science provides the structured lifecycle for transforming raw data into value.

Machine Learning and Deep Learning power predictive systems.

Generative AI creates new content (e.g., text, images, code via GitHub Copilot).

Agentic AI builds autonomous, goal-directed systems.

The future of intelligent systems lies in integrating:

Strong data foundations

Robust modeling techniques

Generative capabilities

Autonomous agent architectures

into scalable, responsible, and domain-aware solutions.

Readings

Primary textbook and selected papers as indicated in class.

Assessment

Final GPA

Contact

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Data Science Diagram

Diagram: [Insert a diagram illustrating Data → Processing → Modeling → Deployment]

Definitions & Related Fields

Business Analytics: The practice of using data analysis and quantitative methods to support business decision-making, often focused on descriptive and diagnostic analytics to improve operations and strategy.

Health Analytics: Application of data analysis, statistical models, and machine learning to health-care data to improve patient outcomes, operational efficiency, and public health decisions.

Business Intelligence (BI): Tools and systems that play a key role in the strategic planning process of a corporation, transforming raw data into meaningful and useful information for business analysis.

Learning Analytics: Measurement, collection, analysis and reporting of data about learners and their contexts, for purposes of understanding and optimizing learning and the environments in which it occurs.

FinTech: Technology-driven innovation in financial services, using data analytics, machine learning, and software to improve financial activities and customer experiences.

Digital Marketing: Use of digital channels and data-driven techniques to reach and engage customers, optimize campaigns, and measure marketing performance.

Relevance to Data Science

Business Analytics: Business analytics overlaps with data science on data cleaning, visualization, and modeling; data scientists provide predictive models while BA focuses on translating insights to business action.

Health Analytics: Health analytics leverages data science methods (EHR analysis, predictive risk modeling) to enable evidence-based clinical decisions and population health management.

Business Intelligence (BI): BI provides dashboards and historical reporting; data science complements BI by adding predictive and prescriptive analytics for forward-looking decisions.

Learning Analytics: Learning analytics uses data science to personalize learning pathways, detect at-risk students, and measure educational interventions.

FinTech: FinTech applies machine learning for credit scoring, fraud detection, and algorithmic trading—areas where data science models are core.

Digital Marketing: Digital marketing relies on data science for customer segmentation, recommendation systems, A/B testing, and campaign optimization.